Practice Set 3 Paper 1 – Basic Sciences in Urology

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Practice paper compiled from the question bank.

Q1. Describe hemodynamic & physiologic changes d/t pneumoperitoneum during lap.

Type: LAQ (25 marks) (also answered, with Q2, in 2019-Summer Paper 1, Q2)

"d/t" is read as "due to" and "lap" as laparoscopy.

Introduction

- **Pneumoperitoneum** is insufflation of gas, usually CO_2 (carbon dioxide), into the peritoneal cavity at 10–15 mmHg to create a working space.
- Its effects have **two drivers**: **pressure** (compression of the cava, renal veins, kidneys and diaphragm) and **absorbed CO_2** (hypercarbia, acidosis, sympathetic drive). Patient position and volume status modify both.

Choice of insufflant

Gas	Advantages	Disadvantages
CO_2 (standard)	Non-combustible; highly blood-soluble, so embolism is less lethal; cheap	Hypercarbia, acidosis, arrhythmia; COPD (chronic obstructive pulmonary disease) patients cannot clear it
Helium, argon	Inert; no hypercarbia	Poorly soluble, so gas embolism risk; costly. Helium used in lung disease or to salvage hypercarbia
N_2O (nitrous oxide)	Less peritoneal irritation	Supports combustion; no electrosurgery
Room air, oxygen	—	Embolism, combustion; abandoned

Effects by pressure (supine, Campbell Table 18.1)

Effect	5 mmHg	10 mmHg	20 mmHg	40 mmHg
Heart rate	↑	↑	↑	↓
MAP (mean arterial pressure), SVR (systemic vascular resistance)	↑	↑	↑	↑
Cardiac output	↔/↓	↔/↑	↔/↓	↓
GFR (glomerular filtration rate), urine output	↔	↓	↓↓	↓↓
Arterial PCO_2	↔	↑	↑	↑
Arterial pH	↔	↔/↓	↓	↓

- Renal function is the first casualty (from 10 mmHg). Afterload rises at every pressure. Output collapses at 40 mmHg. Acidosis builds from 20 mmHg.

Cardiovascular (haemodynamic) changes

- **Afterload rises**: aortic and splanchnic compression plus catecholamine and **vasopressin** release raise MAP and SVR.
- **Preload depends on volume status.** In normovolaemia or hypovolaemia, pressure up to 20 mmHg compresses the IVC (inferior vena cava) and venous return falls. In hypervolaemia the cava resists and venous return may rise. At ≥40 mmHg venous return falls sharply in everyone.
- **Arrhythmias**: hypercapnia causes tachycardia and ventricular ectopics; peritoneal stretch causes **vagal bradycardia** or asystole. A new arrhythmia may warn of pneumothorax,

hypoxia or gas embolism.
- **CVP (central venous pressure) reads falsely high** because intrathoracic pressure rises; do not guide fluids by it.
- Leg venous return falls: theoretical DVT (deep vein thrombosis) risk.

Respiratory changes

- **Diaphragm splinted**: FRC (functional residual capacity) and compliance fall, peak airway pressure rises; atelectasis and V/Q (ventilation–perfusion) mismatch. Dead space is unchanged. PEEP (positive end-expiratory pressure) helps in lung disease.
- **Head-down** adds a further fall in vital capacity, endobronchial tube slip and facial oedema.
- **Barotrauma** (pressure >15 mmHg for long): pneumothorax, pneumomediastinum.

Acid–base changes (CO_2 absorption)

- CO_2 is absorbed through the peritoneum and, extraperitoneally, through fat and disrupted lymphatics. Absorption rises over 30–60 minutes, then plateaus.
- **Respiratory acidosis** and rising end-tidal CO_2. Treat by raising minute ventilation and lowering pressure.
- **COPD**: arterial gases if insufflation exceeds 1 hour, continued in recovery. **Hypercarbia peaks after extubation**, as stored CO_2 is released.

Renal changes

- **Oliguria** comes from **renal vein and parenchymal compression**, not ureteric compression or hormones. Cortical flow falls; medullary flow rises up to 20 mmHg, then falls. It reverses on desufflation.
- Keep pressure ≤10 mmHg. If needed give furosemide, mannitol 12.5–25 g or dopamine 2 µg/kg/min to keep urine >100 mL/hour, rather than fluid boluses (overload, oedema). Important in live donor nephrectomy.

Other systems

- **Splanchnic**: hepatic and mesenteric flow fall (rare mesenteric thrombosis); less ileus than open surgery; no rise in reflux.
- **ICP (intracranial pressure) and IOP (intraocular pressure)** rise with hypercarbia and head-down tilt: caution in head injury, shunts and glaucoma.
- **Stress and immune response**: cortisol, catecholamines and prolactin rise as in any surgery, but hyperglycaemia, insulin resistance, catabolism and IL-6 (interleukin-6) rise less than after open surgery; oncological benefit unproven.
- **Also know:** hypothermia from cold dry gas; shoulder-tip pain from diaphragmatic CO_2 irritation.

Effects of position and approach

Position	Heart rate	MAP	Cardiac output
Head-up	↑	↓	↓
Head-down (Trendelenburg)	↓	↑	↑ (but lungs worse)
Lateral flank, kidney rest	—	—	Lung V/Q mismatch; rhabdomyolysis risk

- **Retroperitoneal** insufflation causes smaller swings in output and venous pressures than transperitoneal; pelvic extraperitoneal insufflation raised arterial CO_2 more.

Safe practice

- Preoperative cardiopulmonary assessment; correct arrhythmias; optimise volume.

- **Work at 10–12 mmHg** (12 mmHg keeps stroke volume unchanged); 20 mmHg only briefly (first trocar, venous bleeding).
- Monitor end-tidal CO_2, saturation, airway pressure and urine output.
- **Recent advances:** low-pressure (8 mmHg) surgery with valveless insufflators (AirSeal); helium for lung disease.

Diagram to draw: abdomen under pressure with arrows to diaphragm (lungs), IVC (preload), aorta (afterload) and renal vein (oliguria), plus CO_2 absorption to blood.

Key points to remember:

- Two drivers: pressure and absorbed CO_2; work at 10–12 mmHg.
- Afterload rises at all pressures; output collapses at 40 mmHg.
- Oliguria from renal vein compression; reversible.
- Hypercarbia peaks after extubation; COPD needs arterial gases.

Q2. Complications of retroperitoneoscopy.

Type: LAQ (25 marks) (also answered, with Q1, in 2019-Summer Paper 1, Q2)

Introduction

- **Retroperitoneoscopy** is laparoscopic or robotic surgery in the retroperitoneum without entering the peritoneum. It gives direct hilar access and avoids bowel, but the space must be **created**.
- **Why its complications differ:** no natural cavity; space made by finger and **balloon dissection** behind Gerota's fascia and in front of the psoas; fewer landmarks; dissection close to the lens; gas tracks along tissue planes; CO_2 (carbon dioxide) is absorbed through fat and lymphatics.
- Minimally invasive renal surgery: about 15% complications, mostly postoperative. Learning curve about 20 cases.

Classification (by phase of surgery)

Phase	Complications
Access and space creation	Wrong plane, balloon rupture, peritoneal tear, muscle bleeding
Insufflation	Subcutaneous emphysema, hypercarbia, pneumothorax, gas embolism
Dissection	Vascular, visceral, nerve, thermal injury
Exit and recovery	Port bleeding, hernia, rhabdomyolysis, urinoma, lymphocele

1. Access and balloon-related

- **Balloon in muscle** → postoperative **flank hernia** or bulge.
- **Too anterior** → peritoneal breach or colon injury; **too posterior** → bleeding from quadratus lumborum or psoas.
- **Balloon rupture**: latex glove balloon breaks into many fragments (silicone leaves one) that must be retrieved.
- **Peritoneal tear**: CO_2 enters the peritoneum and collapses the small working space. Manage by venting with a Veress needle or port, enlarging the tear, or converting to transperitoneal.
- Disorientation: keep the **psoas horizontal** as the landmark.

2. Gas-related

- **Subcutaneous emphysema** (commonest; commoner than transperitoneal): crepitus over flank, chest and neck; pneumoscrotum. Cause: leak around the port. Prevent with a **balloon-tip (blunt-tip) trocar**, tight incisions and 12 rather than 15 mmHg.
- **Hypercarbia**, especially with emphysema; a sealed port removes the difference from transperitoneal surgery.
- **Pneumothorax (1.6–4%, commoner retroperitoneally)**: pleura is near the upper pole and 12th rib. Signs: neck and chest emphysema; tension gives hypotension, absent breath sounds and high airway pressure. Treat with a 16-G needle in the second or third intercostal space, midclavicular line, then chest drain. Pneumomediastinum and pneumopericardium (about 0.8%) also occur.
- **Gas embolism** (gas into an opened renal vein, IVC [inferior vena cava] or lumbar vein): sudden collapse; end-tidal CO_2 rises abruptly, saturation falls, then end-tidal CO_2 falls; "millwheel" murmur. Stop insufflation, **left lateral, head down**, 100% oxygen, aspirate through a central line.

3. Vascular injury

- Commonest complication of minimally invasive urology; major injury <1 in 1000 but fatal in 8–17%. Upper-tract (retroperitoneal-organ) work carries about five times the pelvic vascular risk.
- Sites: renal artery and vein, **lumbar and gonadal veins**, adrenal vein, IVC, aorta. In the small posterior space the IVC or aorta can be mistaken for hilar vessels.
- **Management of venous bleeding**: raise pressure to 25 mmHg, suction to see the site, tamponade with a sponge, haemostatic sealant, extra 5-mm port for suturing or a Satinsky clamp, hand-assist, or convert. Arterial: clip or staple; aortic injury needs conversion. Never clip blindly.

4. Visceral injury

- **Peritoneum and colon**; duodenum (right); spleen, pancreas (left); liver; pleura; ureter; adrenal.
- **Thermal bowel injury** by four mechanisms: direct activation, direct coupling, capacitive coupling, insulation failure. It may present up to 18 days later with low-grade fever and pain; diagnose by CT (computed tomography) with oral contrast.

5. Positioning and nerve injury

- **Rhabdomyolysis** (about 1% retroperitoneal; 0.4% overall): men, BMI (body mass index) ≥25, kidney rest, full flexion, long cases. Brown urine, hip pain, CK (creatine kinase) >5,000 units. Prevent: kidney rest <1 hour, cases <5 hours, avoid hypotension.
- **Nerves**: brachial plexus (axillary roll), sciatic; in the field, **genitofemoral and lateral femoral cutaneous** on the psoas, ilioinguinal and iliohypogastric (thigh numbness, flank bulge).

6. Postoperative

- Bleeding or haematoma; port-site hernia; wound infection.
- **Urinoma** (partial nephrectomy, pyeloplasty); **lymphocele** or chylous leak after hilar or node dissection.
- Delayed bowel injury; DVT (deep vein thrombosis).
- **Conversion** to transperitoneal or open surgery for bleeding, peritoneal tear or lost orientation.

Retroperitoneal versus transperitoneal

Retroperitoneal more	Retroperitoneal less
Emphysema, pneumothorax, hypercarbia, disorientation	Bowel injury, ileus, adhesions; peritoneum spared

Prevention

- Open (Hasson) access below the 12th rib tip; balloon wholly in the retroperitoneum; balloon-tip port; 12 mmHg; trocars under vision; respect the 12th rib and pleura; check haemostasis at low pressure.

Diagram to draw: flank cross-section showing balloon between psoas and Gerota's fascia, with peritoneum pushed medially and port positions.

Key points to remember:

- Space is created, not natural: wrong plane gives hernia, peritoneal tear or bleeding.
- Emphysema commonest; balloon-tip port and 12 mmHg prevent it.
- Pneumothorax 1.6–4%, commoner retroperitoneally.
- Rhabdomyolysis about 1%: limit kidney rest to <1 hour.

Q3a. Clinical trial methodology

Type: SAQ (10 marks)

Definition

- A **clinical trial** is a prospective study that assigns people to an intervention to test its effect on health outcomes. The **RCT (randomised controlled trial)** is the gold standard because randomisation balances known and unknown confounders.

Phases

Phase	Purpose	Subjects
0	Microdose pharmacokinetics	Very few
I	Safety, dose, toxicity	20–80, often volunteers
II	Activity, efficacy signal	Patients, about 100–300
III	Comparative efficacy versus standard	Hundreds to thousands; basis of approval
IV	Post-marketing surveillance	General population

Design

- **Parallel**, **crossover** (own control; stable chronic disease), **factorial** (two questions at once), **cluster** or **adaptive**.
- Hypothesis: **superiority**, **non-inferiority** (pre-set margin) or equivalence.
- Surgical innovation follows the **IDEAL** framework (Idea, Development, Exploration, Assessment, Long-term study).

Randomisation and allocation

- Simple, **block** (equal numbers), **stratified** (by prognostic factors) or **minimisation**.
- **Allocation concealment** (central or web-based) stops selection bias.

Blinding

- Single (patient), double (plus assessor) or triple (plus analyst). Surgery: sham procedures or blinded assessors.

Endpoints

- One **primary endpoint** sets sample size; secondary endpoints are exploratory.
- **Hard** (overall survival) versus **surrogate** endpoints, valid only if they meet the **Prentice criteria**; in prostate cancer only MFS (metastasis-free survival) does.
- **PROs (patient-reported outcomes)** with validated tools; complications graded by **Clavien–Dindo** or CTCAE (Common Terminology Criteria for Adverse Events).

Sample size and analysis

- Set α (type I error, usually 0.05), power (80–90%) and expected effect.
- **ITT (intention-to-treat)** preserves randomisation; per-protocol supports non-inferiority claims.
- Kaplan–Meier curves for time-to-event data.

Ethics and reporting

- Declaration of Helsinki, ICH-GCP (International Council for Harmonisation–Good Clinical Practice), informed consent, ethics committee, DSMB (data safety monitoring board).
- **Registration** before enrolment (CTRI [Clinical Trials Registry–India]).

- **CONSORT** (Consolidated Standards of Reporting Trials): checklist plus **participant flow diagram** (25 items in the 2010 version; updated 2025).
- **Also know:** lead-time bias; Will Rogers phenomenon.

Diagram to draw: CONSORT flow diagram: enrolment → randomisation → allocation → follow-up → analysis.

> **Key points to remember:**
> - Phase III RCT with concealed allocation and blinding is the gold standard.
> - Primary endpoint drives sample size; surrogates need the Prentice criteria.
> - ITT analysis; CONSORT flow diagram for reporting.

Q3b. Classify immunosuppresants. Describe few common protocols.

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 1, Q3f and 2020-Summer Paper 4, Q1)

Taken as immunosuppressants in renal transplantation.

Principle

- T-cell activation needs **signal 1** (TCR [T-cell receptor] → calcineurin → IL-2 [interleukin-2]), **signal 2** (costimulation, CD28–CD80/86) and **signal 3** (IL-2 receptor → mTOR [mammalian target of rapamycin] → proliferation). Drugs block these steps.

Classification

Class	Drugs	Action	Main toxicity
CNI (calcineurin inhibitors)	**Tacrolimus**, ciclosporin	Signal 1	Nephrotoxicity, diabetes, tremor
Antiproliferatives	**Mycophenolate**, azathioprine	Purine synthesis	Diarrhoea, leucopenia
mTOR inhibitors	Sirolimus, everolimus	Signal 3	Lymphocele, poor wound healing
Corticosteroids	Methylprednisolone, prednisolone	Cytokine transcription	Diabetes, bone loss
Costimulation blocker	Belatacept	Signal 2	PTLD (post-transplant lymphoproliferative disorder) if EBV (Epstein–Barr virus)-negative
Depleting antibodies	rATG (rabbit antithymocyte globulin), alemtuzumab	T-cell depletion	Infection, PTLD
Non-depleting antibody	**Basiliximab**	Anti-CD25 (IL-2 receptor)	Few
Antibody/B-cell agents	Rituximab, IVIG (intravenous immunoglobulin), plasmapheresis, eculizumab, imlifidase	Antibody removal or blockade	—

Phases

- **Induction** (peri-transplant), **maintenance** (lifelong), **treatment of rejection**.

Common protocols

1. **Standard-risk (most recipients):** basiliximab 20 mg IV (intravenous) on days 0 and 4 + intraoperative methylprednisolone; then **tacrolimus + mycophenolate + tapering prednisolone**. Over 90% receive induction and tacrolimus–mycophenolate; steroids in more than two-thirds.
2. **High-immunological-risk** (sensitised, retransplant, DGF [delayed graft function] expected): **rATG** induction, then triple therapy.
3. **Steroid-avoidance or early withdrawal:** low-risk recipients, usually after depleting induction.
4. **CNI-sparing:** belatacept- or mTOR-based regimens to spare nephrotoxicity; TRANSFORM: everolimus with reduced CNI non-inferior.

5. **ABO-incompatible or HLA (human leucocyte antigen)-sensitised:** desensitisation with rituximab, plasmapheresis and IVIG; imlifidase for crossmatch-positive.
6. **Rejection:** T-cell-mediated → pulse IV methylprednisolone, rATG for Banff grade II–III; antibody-mediated → steroids, plasmapheresis, IVIG, rituximab.

- **Monitoring:** tacrolimus trough levels (higher early, lower later); CYP3A4 (cytochrome P450 3A4) interactions.
- **Urological relevance:** BK virus ureteric stricture; bladder cancer about 3-fold; lymphocele with mTOR inhibitors.

Diagram to draw: APC (antigen-presenting cell)–T-cell synapse with drugs at signals 1–3.

> **Key points to remember:**
> - Basiliximab or rATG induction; tacrolimus + mycophenolate ± steroids maintenance.
> - CNI nephrotoxic; mTOR inhibitors cause lymphocele.
> - Rejection: steroids for T-cell-mediated; plasmapheresis, IVIG, rituximab for antibody-mediated.

Q3c. Assessment of quality of life after urinary diversion

Type: SAQ (10 marks)

Definition

- **HRQoL (health-related quality of life)**: the patient's view of how disease and treatment affect physical, emotional and social function, plus **bother**. It must be **patient-reported**; doctors underestimate urinary, sexual and bowel dysfunction.

Instruments

Type	Examples	Limitation
Generic	SF-36 (Short Form-36), SF-12, EQ-5D (EuroQol five-dimension)	Insensitive to diversion-specific issues
Cancer-specific	EORTC QLQ-C30 (European Organisation for Research and Treatment of Cancer core questionnaire), FACT-G (Functional Assessment of Cancer Therapy–General)	General cancer domains
Bladder-cancer-specific	**FACT-BI** (40 items), **EORTC QLQ-BLM30** (muscle-invasive module, 30 items), **BCI (Bladder Cancer Index**, 34 items: urinary, bowel, sexual domains), FACT-VCI (Vanderbilt Cystectomy Index)	Weight symptoms of one diversion, so biased in cross-diversion comparisons
Neobladder-specific	IONB-PRO (23 items)	Not usable across diversions

- A good tool is **reliable** (Cronbach $\alpha > 0.7$), **valid** in that population and **responsive** to change.

Domains to assess

- Urinary: continence, catheterisation, stoma problems.
- **Body image**, sexual function, bowel function, social and emotional function, global HRQoL.

Evidence: ileal conduit versus neobladder

- **No diversion proven superior.** Studies are mostly cross-sectional and retrospective, without baseline data or randomisation.
- Verdict depends on instrument: generic tools mostly show **no difference** (largest group of studies); EORTC tools tend to **favour neobladder**; the urinary-focused BCI **penalises neobladder** for leakage.
- **Consistent signal:** body image favours neobladder; urinary function favours conduit.
- Large prospective cohort: global HRQoL exceeded baseline in all diversions by 6 months.
- Neobladder: daytime continence improves over 6–12 months; **night-time incontinence** (up to 50%) is the main bother.
- **Also know:** informed patients have less decision regret; most adapt to their diversion; PROMIS (Patient-Reported Outcomes Measurement Information System).

Practical approach

- Baseline and serial validated questionnaires (3, 6, 12 months), a disease-specific plus a generic tool, and realistic counselling.

Diagram to draw: bar sketch: body image (neobladder better) versus urinary function (conduit better).

Key points to remember:

- Generic (SF-36, EQ-5D) plus bladder-specific (FACT-BI, BLM30, BCI) tools.
- No diversion proven superior; studies flawed.
- Neobladder better body image; conduit better urinary function.

Q3d. Describe evidence based medicine

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 4, Q3f)

Definition

- **EBM (evidence-based medicine)**: "the conscientious, explicit and judicious use of current best evidence in making decisions about the care of individual patients" (**Sackett, 1996**).
- It integrates best research evidence, **clinical expertise** and **patient values**.

Steps ("5 As")

1. **Ask** a focused question in **PICO** form (Population, Intervention, Comparison, Outcome).
2. **Acquire**: PubMed, Cochrane Library, guidelines.
3. **Appraise**: validity (randomisation, blinding), effect size (ARR [absolute risk reduction], **NNT [number needed to treat] = 1/ARR**), precision (95% CI [confidence interval]), applicability.
4. **Apply** with expertise and patient preference.
5. **Assess** (audit) the outcome.

Levels of evidence (Oxford CEBM [Centre for Evidence-Based Medicine], as adapted by EAU [European Association of Urology])

Level	Type of evidence
1a	Systematic review or meta-analysis of RCTs (randomised controlled trials)
1b	At least one RCT
2a	Well-designed controlled study without randomisation
2b	Quasi-experimental (cohort) study
3	Non-experimental: case-control, correlation, case reports
4	Expert committee reports or opinion

- **Grades**: A (consistent level 1), B (level 2–3), C (level 4 or consensus).
- **GRADE** (Grading of Recommendations Assessment, Development and Evaluation): certainty high to very low; RCTs start high, observational studies low; **downgrade** for bias, inconsistency, indirectness, imprecision, publication bias; **upgrade** for large effect or dose–response. Recommendations **strong or weak** (used by EAU). AUA (American Urological Association): evidence strength A/B/C; strong, moderate or conditional recommendations.

Pitfalls in appraisal

- **Bias**: selection, confounding, **lead-time**, **length-time**, stage migration (Will Rogers).
- **Surrogate endpoints** must meet the **Prentice criteria**.
- **Also know:** CONSORT, PRISMA, STROBE reporting checklists; validated PROs (patient-reported outcomes).

Limitations

- Few surgical RCTs; trial patients differ from real ones; publication bias; "cookbook medicine" if patient values are ignored.

Diagram to draw: evidence pyramid and the three overlapping circles of EBM.

Key points to remember:

- Sackett 1996: best evidence + clinical expertise + patient values.
- 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
- Level 1a = systematic review of RCTs; GRADE: strong or weak.
- NNT = 1/ARR; beware lead-time, length-time and confounding bias.

Q3e. What are nanobacteria? Describe their role in urolithiasis.

Type: SAQ (10 marks)

Definition

- **Nanobacteria**, now called **CNPs (calcifying nanoparticles)**, are spherical particles far smaller than bacteria with a **calcium apatite shell**, described in bovine and human serum by **Kajander and Çiftçioglu (1998)** as *Nanobacterium sanguineum*.
- **Their living nature is disputed.** Proposed as the smallest self-replicating bacteria, but later work showed "replication" is mineral nucleation, and that they are **mineralo-protein complexes** (apatite with serum proteins such as fetuin-A). Their bacterial DNA proved to be contamination. Most now regard them as **non-living**.

Claimed properties

Property	Claim
Size	Pass through bacteria-retaining filters
"Growth"	Very slow, in serum-containing medium
Shell	Apatite; resistant to heat and gamma irradiation
Detection	Electron microscopy, von Kossa and Hoechst staining, specific antibodies, culture
Susceptibility	Tetracycline and EDTA (ethylenediaminetetraacetic acid) inhibit them

Proposed role in urolithiasis

- **Nidus theory:** CNPs in blood or urine are filtered or excreted, deposit apatite in renal tubular cells and interstitium, and act as a **nucleating nidus** for crystal growth.
- **Link to Randall plaque:** idiopathic calcium oxalate stones grow on **Randall plaque**, an interstitial **apatite** deposit that starts in the basement membrane of the thin loop of Henle and erodes through the papilla. CNPs are **associated with**, but **not proven to cause**, Randall plaque (Campbell, Chapter 95).
- Early studies found apatite nanoparticles in a large majority of stones; rat studies reported renal calcification after injection.
- Also implicated in atherosclerotic calcification, polycystic kidney cysts and chronic prostatitis.

Therapeutic implications (unproven)

- Tetracycline-based regimens with EDTA were tried in stones and chronic prostatitis/chronic pelvic pain syndrome; no RCT (randomised controlled trial) evidence supports routine use.

Current position

- Not in stone guidelines; pathogenesis is explained by supersaturation, inhibitors, Randall plaque and ductal plugs.

Diagram to draw: renal papilla with interstitial apatite (Randall plaque) eroding through urothelium, CNPs as nidus, and calcium oxalate stone attached.

> **Key points to remember:**
> - Nanobacteria = calcifying nanoparticles; apatite shell; living nature disputed.
> - Proposed nidus for apatite and Randall plaque; association, not causation.
> - Tetracycline/EDTA therapy unproven.

Q3f. Secondary malignancies in Urology.

Type: SAQ (10 marks)

Interpreted as second cancers caused by urological or oncological treatment (radiotherapy, chemotherapy, urinary diversion, immunosuppression).

Definition

- A **second (treatment-related) malignancy** is a new, histologically distinct cancer caused by treatment of a first disease. It appears after a latency of years, often **in the radiation field**.

Causes and risks

Cause	Second cancer	Key facts
Pelvic radiotherapy	**Bladder cancer**, rectal cancer, sarcoma	Bladder RR (relative risk) 2–4 after cervical EBRT (external beam radiotherapy), 1.5–2 after prostate EBRT, smaller after brachytherapy, little with IMRT (intensity-modulated radiotherapy). Latency 15–30 years
Cyclophosphamide	**Bladder cancer**, UTUC (upper tract urothelial carcinoma)	Only chemotherapy proven to cause bladder cancer; metabolites phosphoramide mustard and acrolein. About 4.5-fold, dose-dependent (>20 g cumulative)
Urinary diversion / augmentation	Adenocarcinoma at the uro-intestinal junction	Ureterosigmoidostomy 6–29% after 10–20 years (registry 2.6%); cystoplasty 1.6%; ileal conduit 0.02%. Risk tracks colonic mucosa exposed to urine and faeces
Transplant immunosuppression	Skin cancer, PTLD (post-transplant lymphoproliferative disorder), RCC (renal cell carcinoma), bladder cancer	RCC mostly in native kidneys; bladder cancer about 3-fold and aggressive
Testicular GCT (germ cell tumour) treatment	Solid tumours, leukaemia	60–100% excess after cisplatin or radiotherapy, 200% after both; etoposide-related leukaemia
Paediatric tumours	In-field solid tumours, leukaemia	Wilms tumour 1% at 10 years

Mechanisms

- Radiation and alkylating or topoisomerase-inhibitor chemotherapy: mutagenic DNA damage in normal tissue.
- Immunosuppression: loss of immune surveillance and oncogenic viruses (EBV [Epstein–Barr virus], HPV [human papillomavirus]).
- Diversion: urine–faeces contact at the anastomosis.

Prevention and surveillance

- IMRT, smaller fields; **mesna** with cyclophosphamide (reduces cystitis).
- **Haematuria after pelvic radiotherapy or cyclophosphamide: always cystoscopy.**
- Endoscopy from 5 years after ureterosigmoidostomy; augmentation surveillance from about 10 years; excise a defunctioned ureterosigmoid anastomosis.
- Testis: fewer modalities; low-dose CT (computed tomography) in surveillance.

- **Also know:** pioglitazone, aristolochic acid and phenacetin as drug-related urothelial carcinogens.

Diagram to draw: ureterosigmoid anastomosis showing adenocarcinoma at the junction.

> **Key points to remember:**
> - Pelvic EBRT raises bladder cancer 1.5–4-fold; IMRT little.
> - Cyclophosphamide: only chemotherapy proven to cause bladder cancer.
> - Ureterosigmoidostomy cancer 6–29% after 10–20 years.
> - Transplant: bladder cancer about 3-fold.

Rapid Revision Sheet – Practice Set 3 Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- Afterload rises at all pressures; output collapses at 40 mmHg.
- Oliguria from renal vein compression; reversible.
- Hypercarbia peaks after extubation; COPD needs arterial gases.

Q2. Complications of retroperitoneoscopy. (LAQ)

- Space is created, not natural: wrong plane gives hernia, peritoneal tear or bleeding.
- Emphysema commonest; balloon-tip port and 12 mmHg prevent it.
- Pneumothorax 1.6–4%, commoner retroperitoneally.
- Rhabdomyolysis about 1%: limit kidney rest to <1 hour.

Q3a. Clinical trial methodology (SAQ)

- Phase III RCT with concealed allocation and blinding is the gold standard.
- Primary endpoint drives sample size; surrogates need the Prentice criteria.
- ITT analysis; CONSORT flow diagram for reporting.

Q3b. Classify immunosuppresants. Describe few common protocols. (SAQ)

- Basiliximab or rATG induction; tacrolimus + mycophenolate ± steroids maintenance.
- CNI nephrotoxic; mTOR inhibitors cause lymphocele.
- Rejection: steroids for T-cell-mediated; plasmapheresis, IVIG, rituximab for antibody-mediated.

Q3c. Assessment of quality of life after urinary diversion (SAQ)

- Generic (SF-36, EQ-5D) plus bladder-specific (FACT-BI, BLM30, BCI) tools.
- No diversion proven superior; studies flawed.
- Neobladder better body image; conduit better urinary function.

Q3d. Describe evidence based medicine (SAQ)

- Sackett 1996: best evidence + clinical expertise + patient values.
- 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
- Level 1a = systematic review of RCTs; GRADE: strong or weak.
- NNT = 1/ARR; beware lead-time, length-time and confounding bias.

Q3e. What are nanobacteria? Describe their role in urolithiasis. (SAQ)

- Nanobacteria = calcifying nanoparticles; apatite shell; living nature disputed.
- Proposed nidus for apatite and Randall plaque; association, not causation.
- Tetracycline/EDTA therapy unproven.

Q3f. Secondary malignancies in Urology. (SAQ)

- Pelvic EBRT raises bladder cancer 1.5–4-fold; IMRT little.
- Cyclophosphamide: only chemotherapy proven to cause bladder cancer.
- Ureterosigmoidostomy cancer 6–29% after 10–20 years.
- Transplant: bladder cancer about 3-fold.